

WatchSchedule

Product Vision, Technical Specification & Development Roadmap

Executive Summary

WatchSchedule is a scheduling intelligence platform built specifically for superyacht watchkeeping operations.

Unlike traditional rota software, WatchSchedule is designed to continuously optimise fairness across crew members over time while maintaining operational requirements and schedule continuity.

The long-term vision is to create the industry's first Fairness Operating System™ for watchkeeping operations, combining scheduling automation, fairness intelligence, historical learning and AI-powered decision support.

Product Vision

Current scheduling methods rely on spreadsheets, manual allocation and subjective decision-making.

WatchSchedule aims to:

- Automate watch schedule generation.
- Create measurable fairness.
- Reduce administrative workload.
- Provide transparent scheduling decisions.
- Learn from historical scheduling data.
- Improve fairness over time.
- Become the operational intelligence layer for yacht crew scheduling.

The platform should become more valuable with every schedule generated.

Core Scheduling Platform (MVP)

Crew Management

Requirements

- Crew database
- Rank management
- Department assignment
- Watchkeeper eligibility

- Rotational crew support
 - Relief crew support
 - Joiners and leavers
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Leave Management

Requirements

- Leave requests
 - Leave approval workflow
 - Leave visibility within schedules
 - Leave forecasting
 - Leave impact calculations
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Schedule Generation

Supported Configurations

- Single-watch vessels
- Double-watch vessels
- Triple-watch vessels

Requirements

- Automatic watch allocation
 - Qualification-based assignment
 - Availability-aware allocation
 - Leave-aware allocation
 - Manual override capability
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Schedule Management

Views

- Weekly
- Monthly
- Rotation View

Functions

- Publish schedules
 - Schedule locking
 - Version history
 - Audit trail
 - Captain/OOW approval workflow
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Fairness Engine™

Purpose

The Fairness Engine™ ensures workload is distributed fairly across the crew over time rather than balancing individual schedules in isolation.

Fairness Scoring Architecture

The platform calculates fairness at three levels.

Crew Fairness Score™

Measures fairness experienced by an individual crew member.

Range

0-100

Factors

- Total duties
- Friday duties
- Weekend duties
- Public holiday duties
- Christmas/New Year duties
- Repeated duty assignments
- Consecutive duty risk
- Leave-adjusted availability
- Historical workload distribution

Purpose

Answers:

"How fair has scheduling been to me?"

Schedule Fairness Score™

Measures fairness of the generated schedule as a whole.

Range

0-100

Factors

- Duty distribution
- Friday distribution
- Weekend distribution
- Public holiday distribution
- Christmas/New Year distribution
- Existing Fairness Debt™
- Consecutive duty risk
- Rotation continuity

Purpose

Answers:

"How fair is this schedule?"

Fairness Debt™

Measures accumulated scheduling imbalance.

Generated by:

- Excess Friday duties
- Excess weekend duties
- Excess public holiday duties
- Excess Christmas/New Year duties
- Repeated duty patterns
- Consecutive duty assignments

Purpose

Answers:

"Who should receive priority consideration in future schedules?"

Duty Weighting System

Not all duties carry equal burden.

Default weighting:

Duty Type	Weight
Standard Weekday	1.0
Friday	1.25

Duty Type	Weight
Saturday	1.5
Sunday	1.5
Public Holiday	1.5
Christmas Eve	2.0
Christmas Day	2.5
Boxing Day	2.0
New Year's Eve	2.5
New Year's Day	2.0

All values configurable.

Fairness Components

Watch Balance

- Equal distribution of duties.
- Weighted burden balancing.

Weekend Fairness

- Weekend duties tracked independently.
- Historical balancing.

Public Holiday Fairness

- Holiday tracking.
- Historical balancing.

Christmas & New Year Burden

Dedicated tracking for:

- Christmas Eve
 - Christmas Day
 - Boxing Day
 - New Year's Eve
 - New Year's Day
-

Repeated Duty Check

Detect:

- Repeated Friday allocations.
- Repeated Sunday allocations.
- Repeated holiday allocations.

Apply fairness penalties.

Consecutive Duty Risk

Identify undesirable clustering of duties.

Apply scheduling penalties.

Leave-Aware Fairness™

Approved leave should never disadvantage crew within fairness calculations.

Requirements

- Exclude unavailable crew from fairness calculations.
 - Recalculate fairness based on eligible crew pool.
 - Maintain fairness continuity after leave.
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Leave Impact Score

Measures burden transferred to remaining crew.

Used for:

- Planning
 - Forecasting
 - Schedule optimisation
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Leave Forecasting

Predict future fairness issues caused by planned leave.

Outputs:

- Future staffing gaps
 - Fairness impacts
 - Resource constraints
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Rotation Continuity™

Objective

Maintain predictable scheduling patterns.

Requirements

- Preserve established rotations.
 - Minimise unnecessary changes.
 - Improve schedule stability.
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Rotation Stability Score™

Measures:

- Schedule predictability
 - Pattern consistency
 - Rotation continuity
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Most Due To Serve™

Rank eligible crew based on:

- Crew Fairness Score™
- Fairness Debt™
- Weekend history
- Holiday history
- Christmas history
- Leave-adjusted availability

Outputs:

- Most due for next duty
- Most due for next Friday
- Most due for next weekend
- Most due for next public holiday
- Most due for next Christmas/New Year duty

Fairness Correction Loop™

Purpose

Create a self-correcting scheduling engine.

Every schedule should influence future schedules.

Process

Step 1

Generate schedule.

Step 2

Calculate:

- Crew Fairness Score™
- Schedule Fairness Score™
- Fairness Debt™
- Most Due To Serve™

Step 3

Identify:

- Lowest fairness scores
- Highest fairness debt
- Historical burden concentrations

Step 4

Optimise future schedules.

Goals:

- Improve lowest fairness scores.
 - Reduce fairness debt.
 - Improve overall fairness.
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Explainability Layer

Explain My Schedule™

Crew can view:

- Why they received a duty.
 - Why they received a Friday.
 - Why fairness scores changed.
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Manual Override Tracking

Track:

- AI recommendation
- Human changes
- Fairness impact

Example:

Fairness Score reduced from 92 → 86 following manual edits.

Schedule Health Score™

Operational quality metric.

Measures:

- Coverage gaps
 - Resource shortages
 - Excessive overrides
 - Consecutive duty risk
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Schedule Intelligence Assistant™

Ask the Schedule™

Natural language interface to scheduling data.

Examples:

- Why did Jack receive this duty?
- Who has the highest Fairness Debt™?

- Increase fairness above 90.
 - What happens if Sarah takes leave?
 - Who worked Christmas last year?
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AI Capabilities

Explain Decisions

Provide transparent explanations.

Fairness Analysis

Analyse fairness metrics.

Historical Insights

Query historical schedules.

Operational Queries

Availability, leave and rotation data.

What-If Scenarios

Simulate:

- Leave requests
- Crew changes
- Manual swaps
- Relief crew additions

Schedule Optimisation

Suggest ways to:

- Improve fairness
 - Reduce burden
 - Improve continuity
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Predictive Scheduling Intelligence

Fairness Alerts™

Examples:

- Fairness Score below threshold.
- Excess Friday allocation.

- Growing Fairness Debt™.
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AI Recommendations

Examples:

- Assign next Friday to Jack.
 - Add relief cover in August.
 - Swap duties to improve fairness.
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Scheduling Intelligence Dashboard

Crew Metrics

Display:

- Crew Fairness Score™
 - Fairness Debt™
 - Historical Fairness Score™
 - Total duties
 - Friday duties
 - Weekend duties
 - Public holiday duties
 - Christmas/New Year duties
 - Consecutive duty risk
 - Leave impact
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Schedule Metrics

Display:

- Schedule Fairness Score™
 - Average Crew Fairness Score™
 - Highest Fairness Debt™
 - Lowest Fairness Score™
 - Consecutive Duty Warnings
 - Rotation Stability Score™
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Data & Learning Strategy

Store permanently:

- Generated schedules

- Published schedules
- Manual overrides
- Fairness Scores™
- Fairness Debt™
- Leave events
- AI recommendations
- Scheduling explanations

This dataset forms the foundation of future scheduling intelligence.

Fleet Intelligence Platform (Future)

Fleet Benchmarking™

Compare:

- Fairness Scores
- Rotation Stability
- Duty allocation trends

Across multiple vessels.

Fleet Analytics

Provide:

- Fleet-wide reporting
 - Management insights
 - Operational benchmarking
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Owner & Management Reporting

Monthly reports:

- Fairness trends
 - Duty allocation
 - Holiday distribution
 - Schedule health
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Strategic Moat

Competitors can copy features.

They cannot easily replicate:

- Historical scheduling data
- Manual override learning
- Vessel-specific optimisation
- Fleet benchmarking
- Fairness intelligence history

The longer a vessel uses WatchSchedule, the smarter and more valuable the platform becomes.

Development Priorities

Phase 1 – MVP

- Crew Management
- Leave Management
- Schedule Generation
- Calendar Views
- Publishing Workflow

Phase 2 – Fairness Engine™

- Crew Fairness Score™
- Schedule Fairness Score™
- Fairness Debt™
- Historical Fairness Score™
- Most Due To Serve™

Phase 3 – Fairness Correction Loop™

- Historical learning
- Fairness optimisation
- Self-correcting schedules

Phase 4 – Explainability

- Explain My Schedule™
- Manual Override Tracking
- Schedule Health Score™

Phase 5 – AI Layer

- Ask the Schedule™
- What-if simulator
- AI recommendations

Phase 6 – Predictive Intelligence

- Fairness alerts
- Leave forecasting
- Coverage forecasting

Phase 7 – Analytics

- Reporting
- Fairness timeline
- Management dashboards

Phase 8 – Fleet Intelligence

- Fleet benchmarking
 - Fleet analytics
 - Cross-vessel intelligence
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End State Vision

WatchSchedule becomes the industry standard for watch scheduling and fairness management.

Most scheduling software generates rotas.

WatchSchedule continuously learns, measures fairness, explains decisions, predicts future issues and optimises schedules over time.

The result is a platform that becomes increasingly intelligent, increasingly fair and increasingly difficult to replicate as more scheduling data is accumulated.